Ans. (c) : A fluorescent lamp or fluorescent tube is a gas discharge lamp that uses electricity to excite mercury vapour. The excited mercury atoms produce shortwave ultraviolet light that then causes a phosphor to fluoresce, producing visible light. 6500 K is usually printed on fluorescent tubelight.

66. Consider the following chemicals:

1. Benzene

2. Carbon tetrachoride

3. Sodium carbonate

4. trichloroethylene

Which of the above is/are used as dry cleaning chemicals?

(a) 1 only
(b) 2 only
(c) 1, 2 and 4 only
(d) 1, 2, 3 and 4

Ans. (c) : Dry cleaning is a cleaning process for clothing and textiles involves using a chemical solvents other then water. The solvent used is typically trichloroethylene while carbon tetra chloride and tri chloroethane is used historically. Benzene is other common agent used for this purpose.

67. Consider the following statements:

1. Caffeine, a constituent of tea and coffee is a diuretic.

2. Citric acid is used in soft drinks.

3. Ascorbic acid is essential for the formation of bones and teeth.

4. Citric acid is a good substitution for ascorbic acid in our nutrition.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1, 2 and 3 only
(c) 3 and 4 only
(d) 1, 2, 3 and 4

Ans. (d) : Caffeine also spelled caffei is most commonly consumed by humans in infusions extracted from the bean of the coffee plant and the leaves of the tea bush, as well as from various foods and drinks containing products derived from the kola nut. Acids are added to soft drinks for extra bite and mouth feel. The primary acid used in colas is phosphoric acid, while the one used in citrus flavoured drinks is usually citric acid. Citric acid is a good substitution of ascorbic acid as per nutritional value.

BIOLOGY

68. Consider the following statements:

1. ELISA test is employed as the first and most basic test for an individual to detect cancer.

2. Almost 50% of human beings have Rh+ blood while the remaining have Rh- blood.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (d) : ELISA (Enzyme-Linked Immunosorbent Assay) is an important test for diagnosing viral diseases like AIDS (caused by HIV). Majority of the human beings are Rh+ while very few are Rh–.

69. Consider the following statements with reference to human body:

1. The common bile duct releases its contents into stomach.

2. The pancreatic duct release its contents into duodenum

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (b) : Bile duct and pancreatic duct both releases its contents into duodenum. The digestive enzyme (pancreatic juices) releases to duodenum from pancreatic duct.

70. Which one among the following is not a good source of nutritional calcium?

(a) Rice
(b) Ragi
(c) Skimmed milk
(d) Egg

Ans. (a) :
Rice is not a good source of nutritional calcium. It has only 10 mg calcium in 100 mg of rice.

71. Assertion (A) : Cellulose is used in making shatter-proof glass.

Reason (R) : Polysaccharides are not soluble in water.

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is correct but R is wrong.
(d) A is wrong but R is correct.

Ans. (a) : Cellulose is not soluble in water that is why it is used in making shatter–proof glass. Cellulose is a poly saccharide.

72. What is the name of the vessel that delivers the nutrient rich blood from the stomach and small intestine to the liver?

(a) Left hepatic artery
(b) Hepatic vein
(c) Right hepatic artery
(d) Hepatic portal vein

Ans. (d) : Hepatic portal vein deliver the nutrient rich blood from the stomach and small intestine to the liver. It is the largest gland in the body.

73. Consider the following statements:

1. Meningococcal Meningitis is transmitted from person to person by mosquito bites.

2. Vomiting and neck pain area two of the symptoms of Meningococcal Meningitis.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (b) : Meningococcal meningitis is transmitted from person to person through cough and having a body contact on a regular basis. It is called droplet transmission.

CURRENT AFFAIRS

74. Consider the following statements:

1. Appellate Tribunal for Electricity has been established by each State Government in India.

2. One of the components of the Accelerate Power Development and Reforms Programme (APDRP) is up-gradation of sub-transmission and distribution system for electricity in India.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans. (b) : The Appellate Tribunal for Electricity was constituted in the year 2005 to hear appeals against the orders of the adjudicating officer or the Central and State Electricity Regulatory Commissions under the Electricity Act, 2003. The second appeal lies before the Supreme Court on substantial questions of law only. The Appellate Tribunal for Electricity comprises a Chairperson who has been a Judge of the Supreme Court or Chief Justice of a High Court, one judicial member who has been or qualified to be a judge of High Court, two technical members who are electricity sector experts and one technical member who is an expert from petroleum and natural gas sector. Principal bench of The Appellate Tribunal for Electricity is in Delhi while there are three circuit benches at Kolkata, Mumbai and Delhi. Hence, statement 1 is not correct. Accelerated Power Development Reforms Program (APDRP) is intended for up gradation of sub transmission and distribution in densely electrified zones in the urban and industrial areas and improvement in commercial viability of State Electricity Boards by reducing the aggregate technical and commercial losses to around 15 percent as against the existing over 50 percent. Hence, statement 2 is correct.

75. Consider the following statements in respect of India's advanced satellite INSAT-4A:

1. INSAT-4A was launched in December, 2005 from the Mexico

2. The European Commercial launch service provider Arianespace was associated with the launch of INSAT-4A.

3. Tata Sky- a digital cable service provider is associated with DTH television broadcasting services from INSAT-4A

Which of the statements given above are correct?

(a) 1, 2 and 3
(b) 1 and 2, only
(c) 2 and 3, only
(d) 1 and 3, only

Ans. (c) : INSAT-4A was launched at French Guyana, in December, 2005 by the European Ariane launch vehicle, INSAT-4A which is positioned at 83 E longitude along with GSAT-12, has high power K_u-band transponders with footprint covering Indian mainland and C-band transponders covering Indian geographical boundary and area beyond India in South East and North West regions. Tata Sky- a digital cable service provider is associated with DTH television broadcasting services from INSAT-4A.

76. Consider the following statements:

1. Kofi Annan-the UN Secretary General is from Nigeria.

2. Kofi Annan is the first UN Secretary General to be appointed from the ranks of the United Nations staff.

3. Kofi Annan was appointed for a second term to the office of the UN Secretary General

Which of the statements given above are correct?

(a) 1, 2 and 3
(b) 1 and 2, only
(c) 2 and 3, only
(d) 1 and 2, only

Ans. (c) : The seventh Secretary-General of the United Nations was Kofi Annan of Ghana. As the first Secretary-General to be elected from the ranks of the staff of the United Nations, his term began on 1 January 1997. By acclamation, the General Assembly, acting on a recommendation by the Security Council on 29 June 2001, appointed him to a second term of office, beginning on 1 January 2002 and ending on 31 December 2006.

77. Which one of the following countries is not a participating team in the Football World Cup-2006 to be held in Germany in July, 2006?

(a) Russia
(b) Togo
(c) Ivory Coast
(d) Switzerland

Ans. (a) : The 2006 FIFA World Cup was the 18th FIFA World Cup, the quadrennial international football world championship tournament. It was held from 9 June to 9 July, 2006 in Germany. Russia did not participate in this event.